

Restrictive Versus Permissive Use of Broad-spectrum Antibiotics in Patients Receiving Allogeneic Stem Cell Transplantation and With Early Fever Due to Cytokine Release Syndrome: Evidence for Beneficial Microbiota Protection Without Increase in Infectious Complications

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Background. Intestinal microbiome contributes to the pathophysiology of acute gastrointestinal (GI) graft-versus-host disease (GvHD) and loss of microbiome diversity influences the outcome of patients after allogeneic stem cell transplantation (SCT). Systemic broad-spectrum antibiotics have been identified as a major cause of early intestinal dysbiosis.

Methods. In 2017, our transplant unit at the university hospital in Regensburg changed the antibiotic strategy from a permissive way with initiation of antibiotics in all patients with neutropenic fever independent of the underlying cause and risk to a restrictive use in cases with high likelihood of cytokine release syndrome (eg, after anti-thymocyte globulin [ATG] therapy). We analyzed clinical data and microbiome parameters obtained 7 days after allogeneic SCT from 188 patients with ATG therapy transplanted in 2015/2016 (permissive cohort, n = 101) and 2018/2019 (restrictive cohort, n = 87).

Results. Restrictive antibiotic treatment postponed the beginning of antibiotic administration from 1.4 ± 7.6 days prior to 1.7 ± 5.5 days after SCT ($P = .01$) and significantly reduced the duration of antibiotic administration by 5.8 days ($P < .001$) without increase in infectious complications. Furthermore, we observed beneficial effects of the restrictive strategy compared with the permissive way on microbiome diversity (urinary 3-indoxylsulfate, $P = .01$; Shannon and Simpson indices, $P < .001$) and species abundance 7 days post-transplant as well as a positive trend toward a reduced incidence of severe GI GvHD ($P = .1$).

Conclusions. Our data indicate that microbiota protection can be achieved by a more careful selection of neutropenic patients qualifying for antibiotic treatment during allogeneic SCT without increased risk of infectious complications.

Keywords. allogeneic stem cell transplantation; intestinal microbiome; broad-spectrum antibiotics; outcome.

Acute graft-versus-host disease (GvHD) is still the major cause of morbidity and mortality after T-cell-repleted allogeneic stem cell transplantation (SCT) [1]. In recent years, intestinal microbiota disruption and subsequent immune dysregulation, particularly early after SCT, have been recognized as essential risk factors contributing to the development of gastrointestinal (GI) acute GvHD [2, 3]. The use of broad-spectrum antibiotics for the treatment of neutropenic infections is a major cause of early microbiota damage, which is uniform across transplant

centers worldwide [4–9]. While experimental and some clinical studies show gradual differences in severity of microbiota damage between different broad-spectrum antibiotics [7, 10–13], other clinical trials, including our recent study, revealed similar microbiota damage by almost all currently used antibiotics [14, 15]. Not only the type, but also the timing of antibiotics, plays an important role in intestinal dysbiosis: early start of antibiotic treatment before the day of SCT was associated with more severe microbiota damage and higher GvHD-related mortality compared with antibiotic use on or after the day of SCT [16]. Thus, strategies aiming to postpone initiation of antibiotics might contribute to microbiota protection. While guidelines in the 20th century strongly recommended to treat all patients with severe neutropenia and fever, irrespective of the clinical likelihood of infection (permissive strategy) [17, 18], more recent guidelines propose considering the clinical likelihood of infections and the absence of confounding causes of fever, such as cytokine release syndrome (CRS) (eg, due to

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anti-thymocyte globulin [ATG]) in the indication for starting antibiotics [19–22]. However, clinicians are frequently hesitant to postpone antibiotic treatment due to the potential risk of missing treatment of neutropenic sepsis. Up to 2016, our unit for allogeneic SCT also used a permissive strategy. To balance the risk of microbiota damage by early use of antibiotics and the risk of severe neutropenic infection, in 2017 our unit introduced a defined restrictive antibiotic strategy in recipients with a high likelihood of CRS. Cytokine release syndrome is a very common cause of fever in patients during allogeneic SCT—for example, due to the first-dose effect of ATG, blood products, or the early allo-reaction in haploidentical SCT [23–26]. Recent guidelines recommend consideration of these causes of fever before initiating antibiotic treatment [22]. Our data show significant reduction in antibiotic use and microbiota damage without an increase in infectious complications.

METHODS

Patients

In this retrospective analysis 188 patients undergoing allogeneic SCT at the University Hospital Regensburg were included. A permissive antibiotic strategy ($n = 101$) in patients who underwent transplantation between January 2015 and December 2016 was compared with a more recent cohort of patients who underwent transplantation between January 2018 and December 2019, in which a restrictive antibiotic strategy ($n = 87$) was administered. Patients who underwent transplantation in the interim period of 2017 were excluded to avoid potential bias. In both cohorts, only patients with ATG use were included. ATG is routinely administered for immunosuppression in patients with unrelated donors in a dose of 10 mg/kg body weight per day on day -3 until day -1 and in older patients with sibling donors in a dose of 5 mg/kg body weight per day on days -2 and -1 . Cohorts were comparable except for a higher proportion of unrelated donors in both groups, as explained previously. With approval by the Ethics Committee of the Mount Sinai Acute GVHD International Consortium (MAGIC) (no. 21-2521-101) and after receipt of written informed consent, urinary and fecal samples were collected at day $+7$ (range: day $+2$ to $+10$) to analyze microbiota and clinical outcome parameters. All specimens were stored at -80°C until analysis.

Indication for Initiation of Antibiotic Treatment

Patients from both cohorts received a single gut decontamination with oral rifaximin 200 mg 2 times daily from the start of conditioning until day 30. For antiviral and antifungal prophylaxis, aciclovir 400 mg 4 times per day and azoles were administered. All patients received cotrimoxazole 960 mg 3 times per week to avoid *Pneumocystis jirovecii* pneumonia. Broad-spectrum antibiotics were initiated as soon as patients developed fever higher than 38.3°C and were neutropenic (neutrophils

$<500/\text{mm}^3$) according to international guidelines [19, 20]. Piperacillin/tazobactam at a 3-times-daily dose of 4.0/0.5 g was used as empiric first-line therapy, whereas meropenem 1.0 g 3 times daily and vancomycin 1.0 g twice daily served as second-line therapy. In cases of penicillin intolerance, patients received alternative first-line treatment with ceftazidime. All antibiotics administered between admission (approximately day -7) and discharge from the hospital (approximately day $+28$) were assessed for our analysis. The beginning of antibiotic treatment was always before neutrophil engraftment for all patients.

In the “permissive” cohort, any occurrence of fever without a detectable infectious reason was classified as unexplained fever (UF) and systemic antibiotics were immediately started to not risk any infectious complication during neutropenia. However, the presence of confounding causes such as CRS after infusion of ATG or blood products causative for fever was not considered. In 2017, we changed our guidelines and recommended postponing antibiotics if neutropenic fever occurred in direct temporal relation (<24 h) to infusion of ATG or blood products and if patients had no evidence of severe sepsis or clinically suspected infections, absence of severe pulmonary damage, or any history of preceding neutropenic infections with multidrug-resistant bacteria. In all patients, immediate careful clinical examination and microbiological cultures from blood and urine were mandatory. If patients fulfilled the above criteria, antibiotic treatment was postponed for 24 hours and only initiated in the case of persisting fever or after a next episode of fever without the above-mentioned confounding causes (“restrictive antibiotic strategy”). The indication for restrictive antibiotic use was independent of the presence of neutropenia when fever occurred as long as the above-mentioned criteria were fulfilled.

Outcome Parameters and Analysis

For all patients, the start and duration of treatment with broad-spectrum antibiotics were recorded. Urine samples to detect 3-indoxylsulfate (3-IS) as an indirect parameter of microbiota damage and stool samples to perform 16S rDNA-based microbiome sequencing were collected at day $+7$ after allogeneic SCT and compared between the permissive and restrictive cohorts. Fecal bacterial composition was analyzed using the 454-pyrosequencing technology on a GS FLX platform (Roche Diagnostics) [27], as described previously [14] (permissive cohort), or by high-throughput, semiconductor-based sequencing on an Ion Torrent PGM system (Thermo Fisher Scientific) (restrictive cohort) after amplification of V3 to V4 hypervariable 16S rRNA gene regions from extracted nucleic acids. Although both methods are technologically very similar regarding preparation of the sequencing library and the sequencing process itself, we have implemented several precautions to minimize and control for possible methodological discrepancies—for example, the addition of exogenous spike-in

bacteria into the original stool sample, which served as a process control [28]. All reads were trimmed to identical read length covering the entire V3 to V4 variable regions and rarefied to 10 000 reads per sample to maintain comparability of alpha diversities. Operational taxonomic units were subsequently identified by clustering filtered reads to 99% sequence identity using the vsearch open-source tool for metagenomics. Gini-Simpson indices [29] and Shannon entropies [30] were calculated with the vegan-package 2.3.0 in R (R Foundation for Statistical Computing, Vienna, Austria) [31]. Taxonomy was assigned to each zero-radius Operational Taxonomic Unit (zOTU) using the IDTAXA algorithm of the DECIPHER v2.18 package and the genome taxonomy database (GTDB) release 214 [32]. Linear models for differential abundance analysis (LinDA) were applied to detect significantly different genera between both antibiotic groups using the MicrobiomeStat package [33]. Alpha and log₂ fold-change cutoffs were set to 0.05 and 0.58, respectively, and all zOTUs below a mean relative abundance of 0.1% were used for the analysis.

Urinary 3-IS and creatinine concentrations were analyzed as previously described [3]. To assess the risk of subsequent infectious complications, the occurrence of bacteremia or sepsis due to bacterial blood stream infection, UF, and clinically documented infection was analyzed. Staging and grading of GvHD during the first 100 days after allogeneic SCT were performed according to the MAGIC criteria [1]. For outcome analysis, transplantation-related mortality (TRM), defined as death due to any transplantation-related cause other than disease relapse, and overall survival (OS) to assess the length of time from SCT that patients are still alive were used.

Bioinformatics and Data Analysis

Continuous data are presented as means ($\pm$ standard deviation). Group comparisons were performed using Mann–Whitney *U* tests due to nonnormal data distribution. Absolute and relative frequencies for categorical data were compared between study groups by chi-square tests. All hypotheses were tested with a 2-sided 5% significance level. Kaplan–Meier analysis was performed to assess TRM and OS. IBM SPSS Statistics 28 (IBM Corporation, Armonk, NY) was used for analyses.

RESULTS

Restrictive Strategy Influenced the Beginning and Length of Antibiotic Treatment

Patients' characteristics are summarized in Table 1. A comparison of the 2 cohorts revealed a substantial reduction in the proportion of patients receiving antibiotics prior to the day of allogeneic SCT, from 48.5% (49/101) in the permissive cohort to 25.3% (22/87) in the restrictive cohort ($P = .002$). The restrictive strategy resulted in a delay of initiation of antibiotics from 1.4 ± 7.6 days prior to SCT to 1.7 ± 5.5 days after SCT

Table 1. Patient Characteristics of the Study Group

Antibiotic Strategy	Permissive	Restrictive	<i>P</i>
n	101	87	
Age at SCT, y	52.3 $\pm$ 12.1	54.0 $\pm$ 10.7	.96
Sex (F/M), n/n	35/66	32/55	.76
Underlying disease			
Acute leukemia	61.4%	52.9%	.57
MDS	9.9%	8.0%	
MPS	6.9%	10.3%	
Bone marrow failure	5.0%	3.4%	
Lymphoma/myeloma	16.8%	25.3%	
Stage at SCT			
Early/intermediate	59.4%	64.2%	.51
Advanced	40.6%	45.8%	
Donor			
MRD	5.0%	18.4%	.004
MUD	95.0%	81.6%	
GvHD prophylaxis			
ATG/CNI/MTX	80.2%	66.7%	.05
ATG/CNI/MMF	13.9%	28.7%	
ATG/Other	5.9%	4.6%	
Stem cell source			
Bone marrow	5.9%	12.6%	.06
PBSC	94.1%	87.4%	
Conditioning			
Standard	21.8%	13.8%	.20
RIC	78.2%	86.2%	
Comorbidity score [34]	2.5 $\pm$ 0.8	2.3 $\pm$ 0.9	.42

For continuous data, Mann–Whitney *U* tests and for categorical data chi-square tests were used for statistical testing, respectively.

Abbreviations: ATG, anti-thymocyte globulin; CNI, calcineurin inhibitor; F, female; M, male; MDS, myelodysplastic syndrome; MMF, mycophenolate mofetil; MPS, myeloproliferative syndrome; MRD, matched related donor; MTX, methotrexate; MUD, matched unrelated donor; PBSC, peripheral blood stem cell; RIC, reduced intensity conditioning; SCT, stem cell transplantation.

(mean total delay of 3.1 d; $P = .01$). Similarly, the mean overall duration of antibiotic treatment was reduced by 5.8 days (21.7 ± 12.6 vs 15.9 ± 7.6 d; $P < .001$), indicating that the restrictive strategy did not result in delayed prolonged application of antibiotics (Figure 1). Despite the restrictive use of systemic antibiotics, the proportion of patients without any antibiotic treatment was not higher compared with the permissive group (7% in the restrictive vs 4% in the permissive group; $P =$ not significant [NS]).

Intestinal Microbiome Diversity at Day 7 Post-transplant Is Preserved by the Restrictive Antibiotic Strategy

Urinary 3-IS levels at day 7 after SCT were significantly higher in the restrictive group, indicating better protection of commensal bacteria ($P = .01$). The mean microbiota indices (Shannon and Simpson index) obtained by 16s rRNA sequencing were also substantially higher in restrictive strategy-treated patients (Figure 2), indicating less dysbiosis at day 7 ($P < .001$). A LinDA analysis detected that unfavorable taxa, such as *Enterococcus* spp., *Streptococcus* spp., *Eggerthella lenta*, and *Clostridium innocuum*, were enriched in the permissive

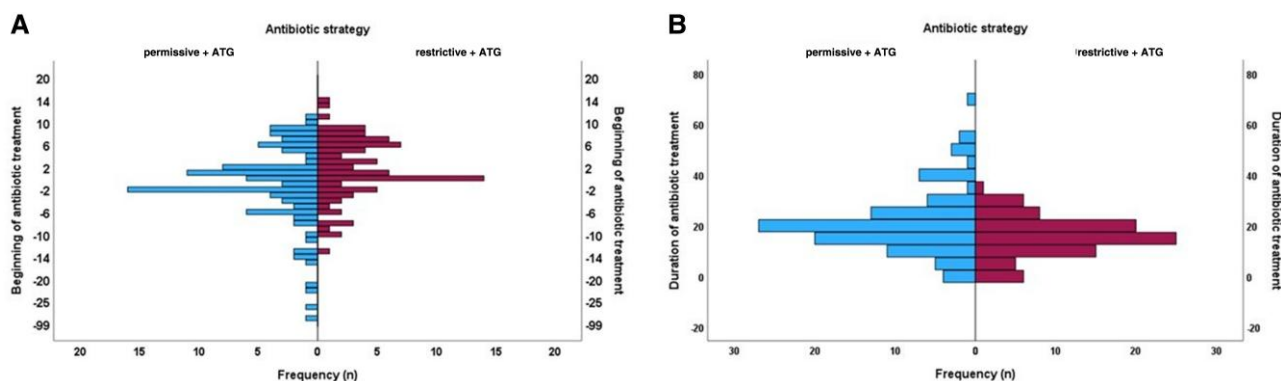


Figure 1. A, Distribution of days of first infusion of broad-spectrum antibiotics in relation to antibiotic strategy. The restrictive antibiotic strategy resulted in a significant delay of initiation of antibiotics ($P = .01$, Mann–Whitney U test). B, The mean duration of treatment with broad-spectrum antibiotics in patients with a permissive compared with a restrictive antibiotic strategy. The restrictive antibiotic strategy was associated with a significant reduction in mean overall duration of antibiotic treatment ($P < .001$, Mann–Whitney U test).

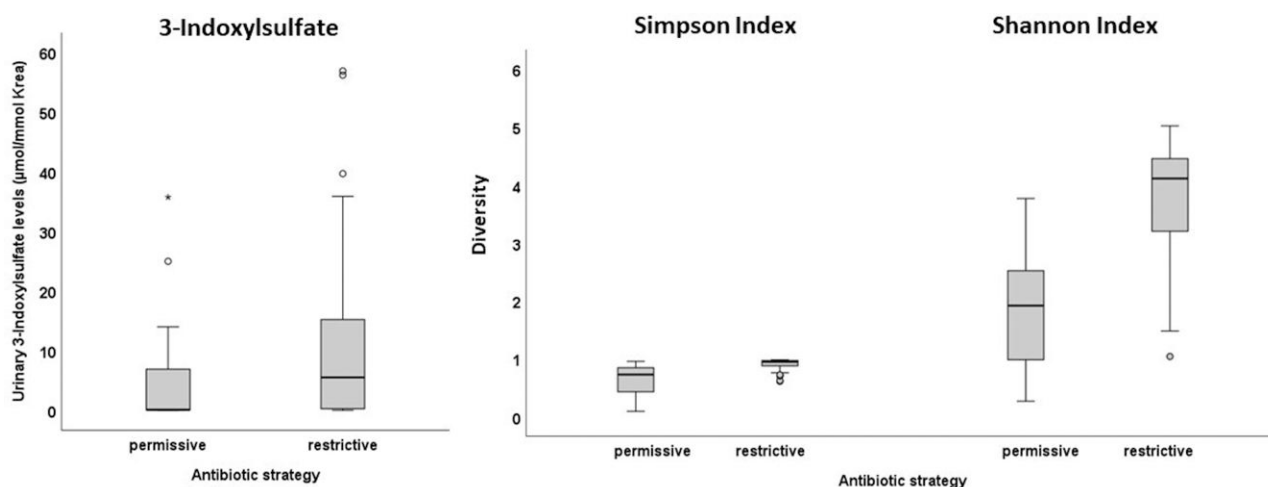


Figure 2. Impact of different types of antibiotic strategies on intestinal microbiota diversity at day 7 after allogeneic SCT as assessed by urinary 3-IS levels as well as 16s rRNA sequencing: 3-IS ($P = .01$) and mean microbiota indices (Shannon and Simpson index; $P < .001$) were significantly higher in the restrictive group. Abbreviations: SCT, stem cell transplantation; 3-IS, 3-indoxylsulfate; Krea, Kreatinine; °, * outliers.

antibiotic group, whereas potentially beneficial taxa, such as *Oscillospiraceae*, *Lachnospiraceae* families, and *Lawsonibacter* genus, were higher in the restrictive antibiotic group. Figure 3 provides an overview of significantly different bacterial species distributed between the 2 antibiotic groups. According to our previous findings [14], we did not detect any protective effects on intestinal microbiome diversity by different antibiotic classes—in particular, piperacillin/tazobactam versus meropenem/vancomycin—in both cohorts.

Restrictive Antibiotic Strategy Was Not Associated With an Increase in Bacterial Sepsis or Documented Infections

A comparison of indications for antibiotic treatment revealed UF as the most frequent indication in both groups, with

48.3% in the restrictive and 36.6% in the permissive cohort ($P = .12$, NS). The incidence of detected bacteremia was slightly higher in the restrictive group (29.9% vs 18.8%; NS), but the incidence of sepsis and any documented infection was even decreased (sepsis: 2.3% vs 8.9% [$P = .05$]; any documented infection: 35.6% vs 52.5% [$P = .02$]) (Table 2).

Restrictive Antibiotic Strategy Resulted in Beneficial Effects on the Incidence of Severe Acute Gastrointestinal Graft-Versus-Host Disease

Although overall GI GvHD was comparable between the 2 cohorts (33.3% vs 32.7%; NS), a trend toward a reduction in severe GvHD stages (grade 2–4) was observed for the restrictive in relation to the permissive group (8.0% vs 15.8%; $P = .1$) (Table 3). Restrictive use of antibiotics was associated

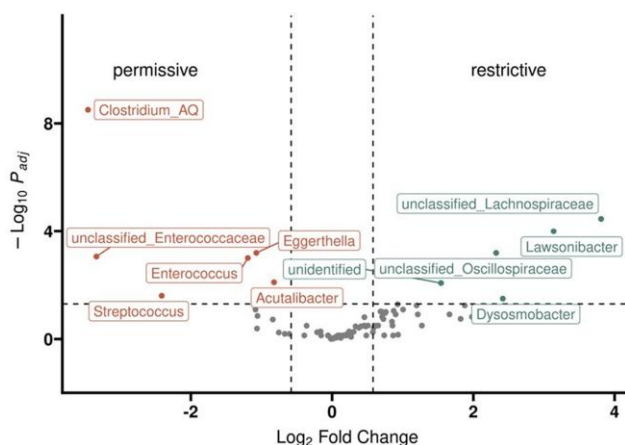


Figure 3. Linear models for differential abundance (LinDA) analysis shows compositional differences between the 2 antibiotic groups. There was an enrichment of unfavorable species in the permissive antibiotic group, whereas potentially beneficial species were more pronounced in the restrictive antibiotic group.

Table 2. Comparison of the Incidence of Infectious Complications Between the 2 Antibiotic Groups

	Permissive	Restrictive	P (Chi-square Test)
Unexplained fever only	36.6% (37/101)	48.3% (42/87)	NS
Bacteremia (overall)	18.8% (19/101)	29.9% (26/87)	NS
Bacterial sepsis	8.9% (9/101)	2.3% (2/87)	.05
Any documented infection	52.5% (53/101)	35.6% (31/87)	.02

Abbreviation: NS, not significant.

with a comparable TRM and OS compared with the permissive use of antibiotics (Figure 4).

DISCUSSION

The description of intestinal dysbiosis as a risk factor for acute GvHD by culture-free sequencing approaches reopened the discussion on the role of antibiotic prophylaxis in the setting of SCT [5, 11]. Former experimental and clinical data clearly stated a beneficial effect of suppression of intestinal microbiota by these approaches on GvHD and mortality [8, 35–37]. However, most trials using the molecular approach indicated that antibiotic prophylaxis resulted in an abundance of single potential pathogenic bacteria rather than complete decontamination and thus might contribute to a worse outcome [38]. A major confounding variable in all of these trials is the additional and even more severe microbiota-damaging effect of broad-spectrum antibiotics used for treatment of neutropenic infections, particularly in heavily pretreated patients with leukemia and after intestinal barrier damage due to conditioning [5, 6, 9]. Delaying antibiotics in

Table 3. Incidence of Acute Intestinal Graft-Versus-Host Disease and Survival After Allogeneic Stem Cell Transplantation in the 2 Antibiotic Groups

	Permissive	Restrictive	P (Chi-square Test)
GI GvHD	32.7% (33/101)	33.3% (29/87)	NS
Severe GI GvHD			
Stage 2–4	15.8% (16/101)	8.0% (7/87)	.10
Overall survival	57.4% (58/101)	64.4% (56/87)	NS
TRM	22.8% (23/101)	14.9% (13/87)	.17

Abbreviations: GI, gastrointestinal; GvHD, graft-versus-host disease; NS, not significant; TRM, transplant-related mortality.

neutropenic infections places patients at increased risk of sepsis and septic shock; thus, empiric antibiotic treatment in patients with UF has become the standard of care [21]. Consequently, the physicians' dilemma is to adequately balance the risk of infection against the risk of microbiota damage [39].

In this context, clinical and experimental data [7, 13] show a different impact of certain classes of antibiotics on microbiota damage favoring fourth-generation cephalosporins over carbapenems and piperacillin/tazobactam, which almost completely destroy commensal bacteria [12]. A randomized trial to address these questions has been initiated [40]. Other studies, including our own, analyzing the effect of different types of antibiotics on global microbiota parameters such as 3-IS or diversity indices (eg, Shannon and Simpson index) revealed no major differences in microbiota damage [14], and similarly, studies outside of allogeneic SCT provided no clear answers [10]. In a cooperative analysis of more than 600 patients, the Memorial Sloan-Kettering Cancer Center (MSKCC) and our group reported timing of antibiotic treatment as an additional risk factor: patients receiving antibiotics prior to the day of SCT have the most severe loss of commensal bacteria and the highest incidence of GvHD and TRM, whereas the outcome was intermediate for those patients receiving antibiotics after the day of SCT and was lowest in patients not requiring additional intravenous antibiotics at all [16]. In the last years, a debate on whether UF should be treated independently from the likely underlying cause has been initiated, and recent guidelines suggest not to treat patients immediately with a high likelihood of fever because of confounding causes such as CRS [22, 40]. Cytokine release syndrome is relatively frequent after chimeric antigen receptor-T (CAR-T) cell treatment [41] but occurs in the setting of an allogeneic SCT under different conditions. With the introduction of pretransplant serotherapy using ATG, CRS due to a first-dose effect of these antibodies on inflammatory cells is a very common cause of fever [23], whereas CRS after transfusion of blood products including stem cells occurs less frequently [24, 25]. Along with the introduction of haploidentical SCT and postponement of immunosuppressive prophylaxis with cyclophosphamide until day 3, early allo-reaction and subsequent expansion of infused donor cells has

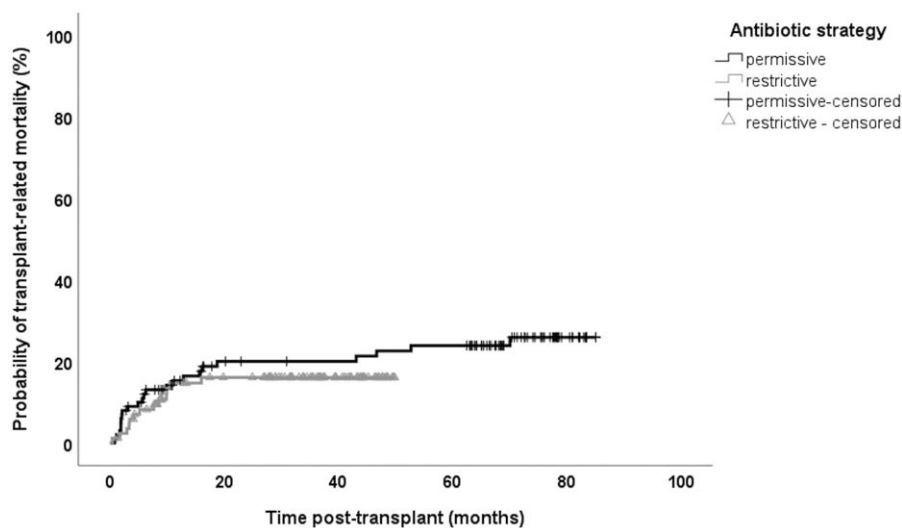


Figure 4. Transplant-related mortality within the first 12 months after allogeneic SCT in relation to antibiotic strategy. The restrictive use of antibiotics was associated with a comparable transplant-related mortality compared with the permissive use of antibiotics (log-rank $P = .4$). Abbreviation: SCT, stem cell transplantation.

become a further frequent cause of CRS [26]. While these causes of UF were neglected in the older guidelines, the more recent guidelines strongly recommend excluding these causes, especially the use of ATG, before initiating antibiotics [22].

Based on these guidelines and our findings on the timing of antibiotics [16, 22], we therefore decided in 2017 to change our institutional guidelines for the treatment of UF and to postpone antibiotics if patients had received treatment inducing CRS and a normal risk of severe infection. We have now compared cohorts of patients receiving antibiotics according to the older permissive strategy with a new cohort treated according to this restrictive strategy. We show that this strategy indeed allows a delay of initiation of antibiotic treatment by 3 days, resulting in a lower proportion of patients receiving antibiotics prior to the day of SCT. Our data indicate that this approach is safe as there was no increase in severe infections, sepsis, or early TRM. However, a strict and careful clinical examination and monitoring of the patient regarding signs of systemic inflammatory response syndrome (SIRS) or sepsis and immediate initiation of an antibiotic therapy whenever a patient's condition worsens are mandatory. Furthermore, the change in practice requires a stringent educational program for clinical staff and the possibility to discuss uncertain cases with senior physicians at any time. The observed slight increase in positive blood cultures can be explained by the possibility that earlier antibiotic treatment in the permissive cohort may have suppressed simple bacterial translocation after mucosal damage. However, this did not translate in an increased incidence of septicemia. On the contrary, the risk of septicemia may even be reduced in the restrictive cohort due to longer maintenance of colonization resistance. Increased protection of commensal

microbiota mediating this effect is also suggested by our direct microbiota analysis, as restrictive-treated patients showed higher 3-IS levels and higher diversity indices on day 7 after SCT. Furthermore, beneficial bacteria, such as *Oscillospiraceae*, *Lachnospiraceae* families [42], and *Lawsonibacter* genus [43], important producers of short-chain fatty acids, were enriched in the restrictive-antibiotics group, whereas potential pathogenic bacteria (eg, *Enterococcus* spp. and *Streptococcus* spp.) discussed to be involved in the pathophysiology of acute GI GvHD [44] were significantly more frequent in the permissive group.

Although our analysis may even indicate a possible beneficial effect of the restrictive strategy on acute GI GvHD development, these results should be interpreted with caution, as our study has several limitations that might contribute to a bias towards lower GvHD incidence. First, this was not a randomized trial. Thus, time-dependent effects may contribute to improved survival, such as improved supportive treatment, changes in the intensity of conditioning, or immunosuppressive prophylaxis. Both cohorts received rifaximin for antibiotic prophylaxis and thus were balanced for this major supportive strategy; similarly, the proportion of patients receiving reduced intensity conditioning (RIC) were comparable. A higher percentage of patients in the restrictive group (18% vs 5%) received grafts from related donors, which could potentially bias our results.

In summary, our data indicate that some microbiota protection can be achieved by a more careful selection of neutropenic patients qualifying for antibiotic treatment in the setting of SCT without putting them at increased risk. Our data strongly support the more recent guidelines on empirical treatment of neutropenic infection [22, 45] and indicate that antibiotic

stewardship should also consider potential microbiota damage whenever possible without risk to the patients, as has already been shown in patients with acute leukemia [46, 47]. Unless better biomarkers indicating the presence of infection at the time of UF are developed, extrapolation of our data on non-SCT patients may be difficult, although similar conditions should apply for CAR-T-cell patients with CRS or CRS occurring before posttransplant cyclophosphamide [26, 48]. If not only the timing of antibiotics and resulting microbiota damage is relevant but also the total duration, additional efforts should be made to apply and analyze the guidelines for de-escalation of antibiotics. Beyond antibiotic timing, novel approaches (eg, the use of selective protection of intestinal microbiota by oral enzymes destroying antibiotics, or absorbents neutralizing antibiotics selectively in the GI tract) may result in additional microbiome protection [49, 50].

Notes

Author Contributions. D. Weber, E. H., E. M., H. P., D. Wolff, W. H., and M. E. were involved in the conception and design of the study; D. Wolff, M. E., and H. P. were responsible for collection of specimens; D. Weber, E. M., and E. H. were responsible for 3-IS measurements; A. H. and A. G. performed bacterial analysis; M. W. contributed to statistical data analysis; B. S. contributed to collection and analysis of infectious data; D. Weber, E. H., E. M., and S.G. collected and analyzed clinical data; D. Weber, M. W., and E. H. wrote the manuscript; and all authors read and corrected the final draft.

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Conflict of interest. E. H. serves on the advisory board for and reports consulting fees from Maat Pharma and Pharmabiom. He also reports payment for educational flyers from Novartis and participation on a Data and Safety Monitoring Board (DSMB) for Medac. D. Wolff declares honoraria from Novartis, Gilead, Takeda, Incyte, Sanofi, and Mallinckrodt and grants or contracts from Novartis; consulting fees from Sanofi and Incyte; support for attending meetings and/or travel from Takeda; participation on a DSMB or advisory board from Behring and Novartis. W. H. declares honoraria from Amgen, Novartis, Janssen Oncology, and Gilead. B. S. reports a role as Chair, German Society of Infectious Diseases (unpaid). D. Weber reports the following grants or contracts unrelated to this work: Deutsche Forschungsgemeinschaft (DFG [German Research Foundation]; project ID 324392634; TRR 221. H. P. reports consulting fees and payment or honoraria for lectures, presentations, speaker's bureaus, manuscript writing, or educational events from Abbvie, Bristol-Myers Squibb, Novartis, Servier, Pfizer, Astellas, and Gilead/Kite; support for attending meetings and/or travel from Abbvie, BMS, Novartis, Servier, Pfizer, Astellas, Amgen, and Jazz; participation on a DSMB or advisory board from Pfizer and Abbvie. M. W. reports grants or contracts unrelated to this work from the German Ministry of Education and Research (13GW0558D). M. E. reports a role as a Directory Board Member from Leibniz Institute for Immunotherapy. None of these declarations are related to the presented work. All other authors report no potential conflicts. All authors have submitted the ICMJE Form for Disclosure of Potential

Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

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